# AI-USE DECLARATION

## Final Exam: AI-Enabled Business Impact Capstone

*AI-use Level 3 analysis declaration and verification record*

### Student details

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| **Student name** | Ayaulym Kenzhebay | **Student ID** | 262687 |
| **Group** | DPA-2602 | **Submission date** | 8 November 2026 |
| **Company A** | Shell | **Company B** | KazMunayGas (KMG) |

### AI-use level and Capstone responsibility

- Capstone analysis operates at AI-use Level 3 (required/designated). Each Level 3 activity must be identified specifically.
- For every AI system used, record what it was used for, what you verified independently, and what you changed as a result.
- You are expected to be able to identify what an AI system got wrong, incomplete or misleading and explain what you did about it.
- You remain accountable for every number, citation and recommendation, regardless of how it was produced.
- The individual defence and live strategic transfer stress test operate at AI-use Level 1: no assistance of any kind.
- Do not upload the OpenStax textbook or the course core text to an AI system.

### A. AI systems register

| AI system / model | Provider / version (if known) | Date(s) used | Capstone purpose(s) |
|---|---|---|---|
| ChatGPT — GPT-6 (OpenAI) | GPT-6; exact session/model setting not separately logged | 18 Oct–9 Nov 2026 (available work period) | Planning and drafting; structuring the KMG/Shell comparison; drafting business context, stakeholder analysis, business model, Human–AI model, operations, marketing, workforce, governance, KPIs, roadmap, financial narrative, presentation wording and website/coursework text. |
| Gemini | Gemini 2.5 Pro (as identified in the draft deck declaration; confirm exact account/version) | Date not recorded in the available project record; enter actual dates if known | Assistance with the initial capstone analysis/deck drafting and synthesis. Outputs treated as drafts, not primary evidence. |
| Python / python-pptx | Python and python-pptx (automation library; not an AI system) | Nov 2026 | Used to edit/format the PowerPoint deck. Listed for transparency but not counted as an AI model. |

### B. Level 3 activity register

Record materially different Level 3 activities across the nine Capstone sections and/or the three graded components. Add rows if needed.

| Section / deliverable | AI system | What I used AI to do | Independent verification | Final decision: accepted / adapted / rejected + change made |
|---|---|---|---|---|
| §1 Business context / company profile | ChatGPT; Gemini | Organise company background, operating context and financial baseline for KMG and Shell. | Compared company claims with KMG and Shell annual reports and investor reporting pages; macro claims checked against World Bank and Kazakhstan statistics. | Adapted: retained only company/context facts traceable to reports; unverified numbers labelled as assumptions or requiring confirmation. |
| §2 Environment & stakeholders | ChatGPT | Structure PESTEL, ownership/stakeholder interests and national energy constraints. | Checked macroeconomic context against World Bank and Bureau of National Statistics; checked legal references against Adilet. | Adapted: removed or flagged legal claims that cited an incorrect/obsolete order number; regulatory wording requires current legal review. |
| §3 Business model / Shell-to-KMG transfer | ChatGPT; Gemini | Compare value chains, business areas and transferability of Shell practices to KMG. | Compared the companies' official annual reports and KMG reporting materials. | Adapted: Shell used as a benchmark, not a model to copy; rejected recommendations inconsistent with KMG's national mandate or verified regulation. |
| §4 Human–AI operating model | ChatGPT | Draft four task categories, human review points, escalation and manual fallback. | Reviewed the proposed workflow for human accountability and consistency with operational safety principles; governance controls remain proposed design, not existing KMG policy. | Adapted: AI positioned as decision support; critical shutdown and safety decisions require qualified human approval. |
| §5a Operations & supply chain | ChatGPT; Gemini | Develop predictive maintenance process, SCADA-to-SAP workflow and proposed operational KPIs. | Compared TUMAR information with KMG Sustainability Report 2025; refinery downtime and telemetry baselines require internal operating records. | Adapted: TUMAR pilot facts retained from KMG reporting; 120-hour downtime and data-quality figures treated as project baselines pending source records. |
| §5b Marketing & customer | ChatGPT | Generate customer journey, fleet portal, | Checked fuel-price claims against Adilet | Adapted: customer-service and inventory ideas |

| Section / deliverable | AI system | What I used AI to do | Independent verification | Final decision: accepted / adapted / rejected + change made |
|---|---|---|---|---|
| | | loyalty and fuel inventory ideas. | legal records; reviewed whether proposed digital services conflict with domestic supply obligations. | retained as proposals; no claim that AI surge pricing is legally permitted. |
| §5c Workforce & career capability | ChatGPT | Structure skills matrix, hiring, retraining and redeployment phases. | Compared workforce and safety context with KMG reporting; proposed hiring/training counts were treated as project assumptions. | Adapted: retraining and human oversight prioritised; headcounts and training hours require management validation. |
| §6 Governance, verification & assurance | ChatGPT | Draft risk register and identify AI-generated claims requiring challenge. | Checked against ISO 31000 and KMG Sustainability Report; legal citation checked against Adilet, including repeal notices. | Adapted: added cyber, bias, reliability, accountability, escalation and manual fallback controls; legal instrument references flagged for correction. |
| §7 Business case & KPIs | ChatGPT; Gemini | Organise CAPEX, NPV, payback, IRR assumptions and KPI dashboard. | Compared available financial figures with KMG annual reporting; arithmetic and underlying cash flows for NPV, IRR, payback and outage savings have not been independently reproduced from a complete model. | Not fully verified: financial outputs retained only as draft estimates/assumptions until the spreadsheet and source records are checked; no invented baseline values added. |
| §8 Roadmap / §9 cross-deliverable integration | ChatGPT | Structure 12-month roadmap, owners, dependencies, gates and consistency across deck/report/video. | Checked roadmap logic against proposed KPI baselines and KMG TUMAR rollout disclosures; confirmed all initiatives are framed as proposals. | Adapted: staged rollout with decision gates retained; targets, budgets and ownership require supervisor/company validation. |

## C. Section-by-section coverage check

Tick where AI was used and reference the corresponding row(s) in Part B. A blank use box means no AI use in that section, not undisclosed use.

| Capstone section | AI used? | Part B row(s) | Notes / verification reference |
|---|---|---|---|
| §1 Business context | ☒ Yes | B1 | Company profile and financial context checked against annual reports. |
| §2 Environment & stakeholders | ☒ Yes | B2 | Macro and legal claims require current-source verification. |
| §3 Business model | ☒ Yes | B3 | Shell benchmark compared with KMG's operating model. |
| §4 Human–AI operating model | ☒ Yes | B4 | Proposed controls; not represented as current KMG policy. |
| §5a Operations & supply chain | ☒ Yes | B5 | TUMAR grounded in KMG report; refinery baselines need internal records. |
| §5b Marketing & customer | ☒ Yes | B6 | Customer ideas treated as proposals; price regulation checked separately. |
| §5c Workforce & career capability | ☒ Yes | B7 | Training and redeployment figures are proposed assumptions. |
| §6 Governance, verification & assurance | ☒ Yes | B8 | Risk controls drafted with reference to ISO 31000 and KMG reporting. |
| §7 Business case & KPIs / §8 Roadmap | ☒ Yes | B9–B10 | Financial model outputs not fully verified; roadmap targets are proposals. |

## D. Business case, KPI and assumption verification (§7)

Every material assumption and number used in the business case should be traceable to evidence and an independent check.

| Assumption / number / KPI | Source or Evidence Folder reference | Independent verification method | Verified result / correction | Where used |
|---|---|---|---|---|
| KMG 2025 production: 26.21 million tonnes (deck figure) | KMG Annual Report 2025 [E1] | Compare with operational results section of report. | Reported production figure must be quoted with the report's exact definition and rounding. | §1 / company context |
| KMG cash, net debt and Net Debt/EBITDA (1,848 bn KZT; 375 bn KZT; 0.16x in draft) | KMG Annual Report 2025 and audited consolidated financial statements [E1], [E2] | Recalculate from audited statements and the company's definition of net debt/EBITDA. | Do not treat draft values as verified until cash restrictions, borrowings and EBITDA definitions are reconciled. | §1 / §7 |
| ANPZ unplanned downtime baseline: 120 hrs/year; target <102 hrs/year | Evidence Folder: internal ANPZ operational logs required; not publicly verified | Obtain 12-month maintenance/outage logs; define planned vs unplanned downtime; recalculate 15% reduction. | 120 hrs/year is a working project baseline only; target is a proposed KPI, not a reported KMG result. | §5a / §7 / KPI dashboard |
| SCADA data quality: 82%; target >98% | Evidence Folder: sensor/SCADA audit required; not publicly verified | Run a data completeness, validity and timeliness audit over a defined period. | 82% is a draft baseline; >98% is a proposed target pending system audit. | §5a / KPI dashboard |
| ABAI expansion: 0 project cells baseline; 1,660-cell target | KMG official reporting and project evidence required [E1], [E2] | Confirm whether 1,660 is the correct number of cells and define whether 0 means new-project deployment, not all existing company activity. | Use '0 deployed under this proposed project scope' only if supported by the project scope; do not imply KMG has no existing ABAI activity. | §5a / §7 / roadmap |
| CAPEX 375m KZT; NPV 1.42bn KZT; payback 7.4 months; IRR 42% | Draft business-case model; underlying workbook required | Rebuild cash flows, timing, tax, discount rate, working capital, residual value and sensitivity cases. | Not independently verified. Label as estimates and do not submit as audited/confirmed returns without a reproducible model. | §7 |
| TUMAR pilot: 10 crews; planned rollout across 60 crews | KMG Sustainability Report 2025 [E2] | Compare the report's description and dates with the deck wording. | KMG report supports a January 2025 pilot on 10 crews and planned deployment to 60 crews; distinguish completed pilot from planned rollout. | §5a / §6 / roadmap |
| Retail fuel price regulation / 'Order No. 171' in draft | Adilet legal records [E8], [E9] | Check exact instrument number, effective date and repeal status in Adilet. | The cited 'Order No. 171' was not substantiated. Historical price-cap instruments identified in the research were repealed in 2025; current legal position must be checked before making a categorical claim. | §2 / §5b / §6 |

### E. AI error, disagreement and correction log

Record at least one material example where an AI output was wrong, unsupported, incomplete, misleading or inconsistent with your evidence. Add rows for additional examples.

| AI system / activity | What was wrong or weak | How I detected it | Evidence used | Correction and effect on final recommendation |
|---|---|---|---|---|
| ChatGPT / legal and pricing analysis | Draft deck referred to 'Order No. 171' as the legal basis for retail price caps; number and current status were unsupported. | Searched the official Adilet legal database and checked the cited orders' status and repeal information. | Adilet records for Orders No. 361, 139/140 and Ministry Order No. 43-н/қ [E8], [E9]. | State only that fuel-price regulation must be checked against the currently effective legal framework; obtain legal confirmation before asserting a current cap. |
| AI-assisted financial narrative | Draft ratios and returns could be presented as settled facts even though definitions and cash-flow calculations were not fully available. | Compared the figures with KMG's official reporting and identified the need to reconcile debt, cash, EBITDA and cash-flow assumptions. | KMG Annual Report 2025 and audited financial statements [E1], [E3]. | Financial values such as 0.16x, 1.42bn KZT NPV, 7.4-month payback and 42% IRR are marked for independent recalculation; no unsupported verification claim is made. |
| AI-assisted operations section | The draft risked presenting 120 hours/year downtime and 82% SCADA data quality as publicly verified KMG facts. | Checked public KMG reporting; no public source in the current evidence set substantiates these exact baselines. | KMG Annual Report and Sustainability Report [E1], [E2]; internal ANPZ/SCADA records still required. | Keep as proposed working baselines only until source logs are obtained; targets are proposals. |
| AI-assisted project transfer analysis | A Shell practice could be transferred directly without sufficient consideration of KMG's national and regulatory context. | Compared KMG and Shell official annual reports and KMG's operating/sustainability disclosures. | KMG Annual Report 2025 [E1]; Shell Annual Report 2025 [E4]. | Changed the recommendation to selective adoption/adaptation/pilot; do not copy Shell's model wholesale. |

### F. AI-generated / AI-edited visual and media disclosure

Complete for visuals or media used in the Deck, Report or Video. AI-generated visuals must be labelled on the visual itself.

| Visual / media + location | AI system | Generated / edited / other | Labelled on visual? Yes / No | Verification / attribution notes |
|---|---|---|---|---|
| PowerPoint charts, tables, process diagrams and KPI dashboard | ChatGPT / Gemini assistance; PowerPoint and python-pptx for layout | AI-assisted drafting/layout; data visualisations and diagrams are proposed/author-edited | No AI-generated photographic media confirmed in the available record; inspect final deck and label any generated images if present. | Figures must be attributed to the relevant source or labelled as project estimates/proposals. |
| Company / refinery photographs or logos, if included | Source-specific; not established in the available record | Externally sourced / inserted (confirm actual image provenance) | To be confirmed in final deck. | Add source/credit near each image; do not describe AI-generated imagery as a real KMG or Shell facility. |

### G. Cross-deliverable consistency and judgment check

+The Report, Deck and Video state the same core recommendation for Company B.

+ Any AI-supported content carried across components was independently rechecked for consistency.

+ The 12-month roadmap is based on my evidence and contains action, owner, KPI baseline/target, dependency, cost basis, risk/control and decision point for each initiative.

+ The governance position covers privacy, security, bias, reliability, accountability, escalation and fallback across the proposed model.

+ All sources used in the Capstone are included in the Evidence Folder and the IEEE references are checked.

### H. Capstone reflection on AI-supported judgment

**The most important AI-supported contribution to my analysis was helping organise a complex comparison between Shell and KMG into a practical strategy covering operations, governance, workforce, KPIs and a 12-month implementation roadmap. AI was used to structure and improve drafts, not as a substitute for company evidence.**

**The most important AI error or limitation was the risk of presenting unsupported legal and financial claims as confirmed facts. In particular, the draft's reference to 'Order No. 171' was not supported by the legal sources checked, and several financial and operational assumptions still require underlying calculations or internal company records.**

**After verification, I changed the recommendation from copying Shell's practices to selectively adopting, adapting, piloting or rejecting them according to KMG's context. I also separated historical company disclosures from proposed targets and working assumptions, and flagged the legal reference and financial model for further checking.**

**The main uncertainties are the exact current legal basis for regulated fuel prices, the internal ANPZ downtime and SCADA data baselines, the scope of the ABAI 1,660-cell target, and the cash-flow model supporting NPV, IRR and payback. Before committing, I would need current legal advice, company operating records and a reproducible financial model.**

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*Completion note: Student ID, exact Gemini use dates/version, actual image provenance and final verification of the business-case workbook must be completed by the student before signing. The checkboxes in Parts G and I and the signature/date fields are intentionally left for the student.*

### I. Defence / live stress-test acknowledgement

+ I understand that AI-use Level 1 applies throughout the individual defence and live strategic transfer stress test.

+ I will answer defence questions from my own knowledge of the submitted work and evidence.

**Optional post-defence confirmation (if required by the instructor):**

☐ I confirm that I used no AI assistance during the defence and live stress test.

### Final declaration

I declare that this form accurately records my AI use across the Capstone. I independently verified the claims, numbers, sources and assumptions for which I remain accountable; I identified Level 3 activities specifically; I recorded material AI errors or weaknesses that affected my work; and the final recommendations and judgments are my own. I understand that AI assistance is prohibited during the individual defence and live strategic transfer stress test.

+ I confirm that every AI system I used is listed in this declaration.

+ I confirm that I did not fabricate, invent or knowingly submit unverifiable data, claims or references.

+ I confirm that I did not upload the OpenStax textbook or the course core text to an AI system.

+ I understand that the defence and live stress test operate at AI-use Level 1: no AI assistance of any kind.

**Student signature: Kenzhebay Ayaulym  Date: 08.11.2026**